

Release checklist

Repository source: RELEASE_CHECKLIST.md

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Release version: v1.0.0 **Release date:** September 26, 2026 **Checklist updated:** September 26, 2026 **Audited pre-release commit:** 892d4bc6fad07950e78da66e45b95063a6415af6 **Audited hosted workflow:** <https://github.com/swihart/minvol-degree3-contact-bound/actions/runs/35468019663>

A checked item means that evidence exists in the current construction checkpoint. It does not authorize publication by itself.

A. Mathematical claim

- ☒ Exact theorem numerator and denominator are fixed in the authoritative certificate.
- ☒ Lower and near-Jung circumradius branches cover the full Jung interval.
- ☒ 03B02 is the exact lower-branch bottleneck.
- ☒ Near-Jung handoff margin is strictly positive in exact arithmetic.
- ☒ Width scaling restores the factor d^3 .
- ☒ Paper states no sharpness, equality, minimizer, or Meissner-extremality claim.
- ☐ Independent subject-matter reviewer has checked the mathematical reduction.

The unchecked external-review item is not required to label the release as a non-peer-reviewed research preprint, but its absence must remain prominent.

B. Exact certificate and audits

- ☒ Standalone verifier has no private-repository dependency.
- ☒ Immutable proof objects have SHA-256 fingerprints.
- ☒ Exact certificate reconstructs byte identically.
- ☒ Independent binary64 Python audit passes.
- ☒ Independent base-R audit passes on the named author's machine.
- ☒ Independent base-R audit passes in hosted R 4.6.1.
- ☒ Ten reconstruction/regression tests pass.
- ☒ Seven adversarial mutations are rejected.
- ☒ Export-equivalence record checks ten theorem-critical fields.
- ☒ Complete replay passes from a fresh clone of the staging remote at commit 892d4bc6fad07950e78da66e45b95063a6415af6.
- ☐ Complete replay passes from a fresh checkout of the immutable v1.0.0 tag.

C. Paper

- ☒ Focused Paper v1 contains only the proof supporting this coefficient.
- ☒ The theorem coefficient and proof-critical constants are generated from the certificate.
- ☒ Paper/certificate reconciliation passes.
- ☒ Reference PDF builds and has been visually inspected page by page during construction.
- ☒ Table 3 row spacing was repaired and approved by the named author.
- ☒ Paper limitations and computer-assisted status are explicit.
- ☒ Sole author and author order approved: Bruce J. Swihart.
- ☒ No institutional affiliation is asserted; correspondence is directed to the repository issue tracker; no ORCID is asserted.
- ☒ Version, date, canonical repository, and versioned release URL inserted.
- ☐ Final ten-page PDF approved in full by the named author for publication.

D. Documentation

- ☒ Reproducibility guide.
- ☒ Verification-status page.
- ☒ Trust-boundary document.
- ☒ Claims-and-limitations document.
- ☒ Certificate specification.
- ☒ Independent-audit record.
- ☒ Mathematical overview.
- ☒ Claim-dependency graph.
- ☒ Proof ledger and internal proof audit.
- ☒ Source ledger with dated literature check.
- ☒ Exact-constants appendix generated from the certificate.
- ☒ Certificate-architecture appendix.
- ☒ Replay-transcript appendix.
- ☒ AI-assistance and provenance disclosure.
- ☒ Clean-source, hosted, and fresh-clone record.
- ☒ Release notes and release checklist.
- ☒ Every Markdown source has a checksummed PDF counterpart.
- ☒ Remote evidence is indexed in `certificate/evidence/rc1_remote_audit.json`.
- ☒ GitHub README math syntax uses supported `$$` and `\mathrm` forms.
- ☒ Named author approved the README after hosted rendering repair.
- ☐ Final release notes and link review approved by the named author.

E. Metadata and legal

- ☒ Repository name confirmed: `swihart/minvol-degree3-contact-bound`.
- ☒ Repository description confirmed.
- ☒ Split license selected and `LICENSE` added: CC BY 4.0 for paper/prose and MIT for software/build infrastructure.
- ☒ `CITATION.cff` added and checked by the metadata preflight.
- ☒ Preferred paper citation finalized for version `v1.0.0`.
- ☒ `RELEASE_DATE` is `2026-09-26`.
- ☒ `VERSION` is `v1.0.0`.
- ☐ DOI or archival identifier added only after one exists.
- ☒ Copyright and third-party license boundary reviewed and documented.

The unchecked DOI item is informational; no DOI is required for the initial GitHub release, and none is implied.

F. Continuous integration and portability

- ☒ Workflow has separate Python, base-R, and document jobs.
- ☒ Hosted document job installs Poppler and checks fonts, text, and page validity.
- ☒ Local Mac without Poppler is documented rather than treated as a theorem failure.
- ☒ PDF byte identity across platforms is not required.

- ☒ Three hosted jobs are green on audited pre-release commit `892d4bc6fad07950e78da66e45b95063a6415af6`.
- ☒ The successful pre-release run is recorded at <https://github.com/swihart/minvol-degree3-contact-bound/actions/runs/35468019663>.
- ☒ A fresh clone of that commit completed the Python and base-R replay, rebuilt the paper and 22 documentation PDFs, and ended clean.
- ☒ Three hosted jobs are green on the rendering-repair/evidence commit, as reported by the named author.
- ☐ Three hosted jobs are green on the final release commit.
- ☐ Three hosted jobs are green on the immutable `v1.0.0` tag.
- ☐ Clean tagged-checkout replay is completed and archived.

G. Release assets

- ☒ Deterministic release-asset builder and verifier are included.
- ☒ Preview paper PDF asset builds and verifies.
- ☒ Preview paper-source archive builds and verifies.
- ☒ Preview standalone certificate archive builds and verifies.
- ☒ Preview documentation archive builds and verifies.
- ☒ Preview replay-transcript archive builds and verifies.
- ☒ Preview release record and one asset `SHA256SUMS.txt` build and verify.
- ☒ Release notes PDF is generated from the final Markdown source.
- ☐ All assets are rebuilt from the exact `v1.0.0` tagged checkout.
- ☐ Every downloaded tagged asset is reverified in a clean directory.

H. Public wording and approval

- ☒ Wording says “computer-assisted” and “not peer reviewed.”
- ☒ Wording does not say “sharp,” “optimal,” “proved Meissner,” or “independently certified.”
- ☒ [Earlier public MinVol spectral repository](#) is clearly separate.
- ☒ Unreleased research results and private history are absent.
- ☒ Named author approved creation and use of the private staging remote.
- ☒ Named author approves the README.
- ☒ Named author approves the theorem wording as displayed in the README.
- ☐ Named author approves the final PDF in full.
- ☐ Named author approves the final release notes.
- ☐ Named author approves the announcement text.
- ☐ Named author explicitly approves changing repository visibility to public.
- ☐ Named author explicitly approves creation of the GitHub Release.

Release decision

Current decision:

PREPARED FOR THE FINAL TAG GATE - DO NOT PUBLISH YET. Version/date meta-data, final README rendering, Table 3 spacing, exact verification, documentation, release tooling, and pre-release reproducibility evidence are prepared. Publication still requires green final-commit and tag workflows, a clean tagged replay, tagged-asset verification, final PDF/release-note/ announcement approval, and explicit approval of the visibility change and GitHub Release.